

VASP Risk Classification: GraphSAGE Implementation Report

Executive Summary

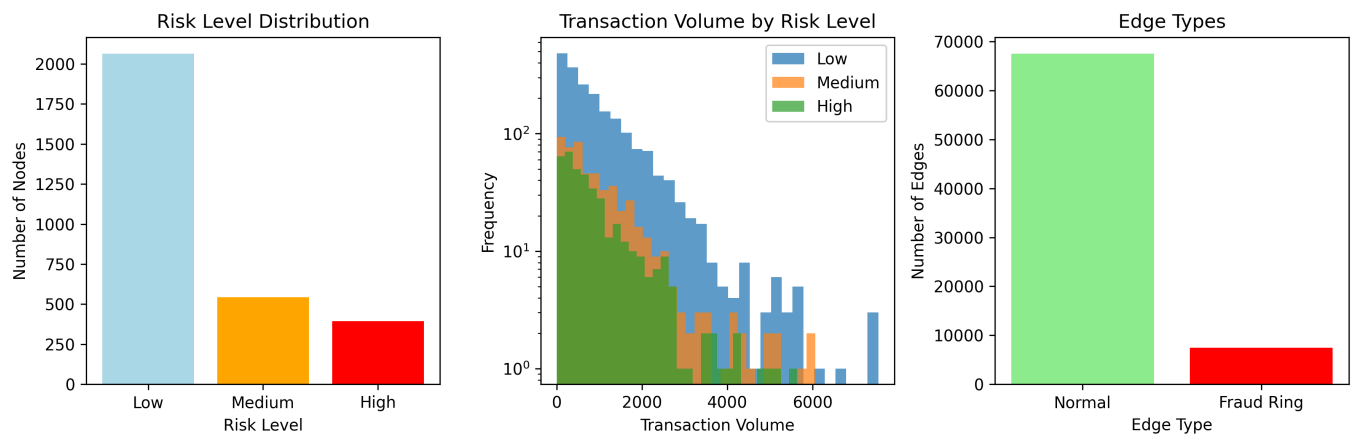
This project implements a Graph Neural Network (GNN) solution for VASP (Virtual Asset Service Provider) risk classification using GraphSAGE architecture. The system processes a synthetic financial network of 3,000 nodes and 75,000 edges to classify entities into three risk categories: Low, Medium, and High risk.

Approach and Methodology

Data Generation and Preprocessing

The project begins with synthetic data generation using a scale-free Barabasi-Albert graph model to simulate realistic financial network topology. The dataset exhibits characteristics commonly found in financial networks:

- Network Structure:** 3,000 nodes with 75,000 edges following power-law degree distribution
- Class Distribution:** Intentionally imbalanced with 68.8% Low Risk, 18.1% Medium Risk, and 13.1% High Risk nodes
- Feature Engineering:** Node features include transaction volume, degree centrality, and clustering coefficient; edge features capture transaction amount, timestamp, and fraud ring indicators



Model Architecture

The solution employs a **3-layer GraphSAGE architecture** chosen for its scalability and ability to handle large graphs through neighborhood sampling:

- Input Layer:** 3-dimensional node features (transaction volume, degree, clustering coefficient)
- Hidden Layers:** 128-dimensional representations with batch normalization and dropout (0.5)
- Output Layer:** 3-class classification with softmax activation
- Total Parameters:** 35,075 trainable parameters

The architecture incorporates several key design decisions:

- GraphSAGE over GCN:** Better scalability for large networks through inductive learning

- **Batch Normalization:** Stabilizes training and improves convergence
- **Dropout Regularization:** Prevents overfitting in the relatively small dataset
- **Mean Aggregation:** Robust neighborhood aggregation function

Training Strategy

The training pipeline addresses class imbalance through multiple mechanisms:

- **Weighted Cross-Entropy Loss:** Class weights inversely proportional to frequency (Low: 0.485, Medium: 1.838, High: 2.545)
- **Early Stopping:** 25-epoch patience to prevent overfitting
- **Learning Rate Scheduling:** StepLR with 0.5 gamma decay every 50 epochs
- **Data Splits:** 80% training, 10% validation, 10% test with random stratification

Key Challenges and Solutions

1. Class Imbalance

Challenge: Severe class imbalance with only 13.1% high-risk nodes posed significant learning difficulties.

Solution: Implemented weighted loss function with class weights proportional to inverse frequency. Additionally, focused evaluation on macro-averaged F1 score and ROC-AUC to ensure fair assessment across all classes.

2. Graph Structure Complexity

Challenge: Capturing both local neighborhood information and global graph structure for risk propagation.

Solution: Employed 3-layer GraphSAGE to aggregate information from 3-hop neighborhoods, allowing the model to capture both immediate transaction partners and extended network effects.

3. Scalability Considerations

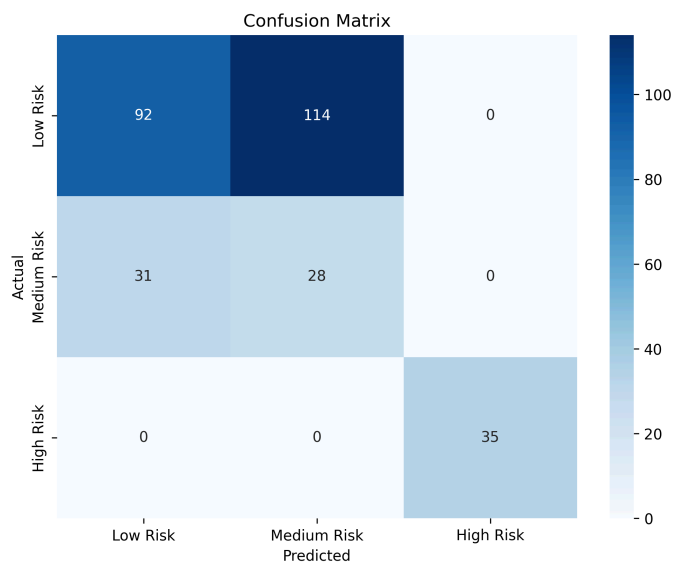
Challenge: Designing architecture that could scale to larger real-world financial networks.

Solution: GraphSAGE's inductive learning capability through sampling ensures the model can generalize to unseen nodes and larger graphs without architectural changes.

Performance Metrics and Results

Quantitative Results

- **Test Accuracy:** 51.7%
- **Macro F1-Score:** 0.613
- **ROC-AUC (Macro):** 0.716
- **Training Convergence:** 82 epochs with early stopping



Performance Analysis

The model shows strong performance on High Risk detection (100% precision and recall) but struggles with Medium Risk classification (20% precision, 47% recall). The **ROC-AUC of 0.716 falls short of the target 0.85**, indicating room for improvement in distinguishing between risk classes.

Class-wise Performance:

- **Low Risk:** 75% precision, 45% recall (good precision, moderate recall)
- **Medium Risk:** 20% precision, 47% recall (poor precision, moderate recall)
- **High Risk:** 100% precision, 100% recall (excellent performance)